Final Report

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1 Abstract

In the following project we study a randomized clinical trial of 134 laboratory animals that start having an administration of different drug doses per day. This drug tries to improve their bone quality, so at a certain time (4 or 7 weeks), the study is ended and the bone characteristics are measured.

After an organization of the given database and dealing with its common challenges (NA observations, outliers,...) we have performed an extensive statistic analysis using different usual multivariate techniques.

Firstly, we applied PCA and Factor Analysis and they let us satisfactorily reduce the dimension of the problem with the aim of grouping variables related to bone quality in three new ones. And in this way, being able to better study the effect that initial weight, time and dose have on these.

Finally, the results that provide us the Discriminant Analysis show us in a clear way that high doses produce toxic effects to the individuals, and small doses seem to slightly help the bone quality. Anyway, the drug seems not to be very effective.

2 Introduction

We have been given a randomized clinical trial of 134 laboratory animals. At 2-weeks age they start having an administration of different drug doses per day: 1, 8, 10, 80, 100, 400, 500, 1000 mg/day and placebo.

We divided the measurements in 3 steps: initial (0 weeks), intermediate (2 weeks, study 1 and 3; 4 weeks study 2) and final (4 weeks, study 1 and 3; 7 weeks study 2). Only the weight can be measured at the intermediate step whereas the primary bone variables (Tb.Sp, Tb.N, BMC (mg), BMD (mg/cm^2), length (mm), GPSurface (mm^2), GPVolume (mm^3)) can obviously only be taken at the last step.

Now some questions are presented to us:

1. How does initial weight (weight at the start of treatment) influence bone quality?
2. Does the administration of different doses across the three studies have a significant effect?
3. To what extent does treatment duration (longer or shorter) affect bone quality?
4. How can minimum and maximum dose ranges be calibrated as a function of time to ensure clinically meaningful bone quality?

Following this indications we have been given a database with all the measurements and we decided to organize it in a more comfortable way for applying statistics methods.

In order to answer the previous questions we used 3 different multivariate techniques with two purposes:

1. **PCA and Factor Analysis:** to reduce the dimension of the problem, i.e group the variables in a coherent way to study more easily the problem.
2. **Factor Analysis:** that provide us useful information and graphics that let us clarify the solution.

3 Materials and Methods

3.1 Materials

This subsection will be dedicated to describe the data base. In the following, we will name and explain what is the meaning of each variable of the original database:

1. *Grupo*: It gives the name and the number of the group where each rat belongs to.
2. *Id*: It is just the number that identifies each individual.
3. *Estudio*: It shows which study each rat belongs to. It goes from 1 to 3.
4. *Dosis*: It is the amount of the drug that will be given to the individuals. Its values are 0 (placebo), 1, 10, 80, 100, 400, 500 and 1000 mg/day.

5. *Tiempo*: It is the time periods of interest of the experiment. Its values are 0, 14, 28 and 49 days. Depending of which study we look at, 28 and 49 days are the durations of the experiment.
6. *Peso*: It is the weight (grams) of each rat. It variates during the passage of time.
7. *Tb.Sp*: It represents the space between the trabeculae of the tibia.
8. *Tb.N*: It represents the number of trabeculae of the tibia.
9. *BMC*: Bone Mineral Content.
10. *BMD*: Bone Mineral Density.
11. *Longitud*: It measures the lenght of the tibia.
12. *GPSurface*: Growth Plate Surface.
13. *GPVolume*: Growth Plate Volume.

However, since the way that the database was distributed gave us problems when using our software tools, we made the decision of dividing the variable *Peso* in three new and different variables:

1. *Peso0*: The initial weight of each rat before the experiment.
2. *Peso1*: The weight of each rat in the middle of the experiment (14 or 28 days, depending of the studio).
3. *Peso2*: The final weight of each rat when all the other measures are taken.

We also create a new variable called "28/49" wheter the duration of the experiment was 28 or 49 days, giving a 0 or a 1 respectively.

Table 1: Basic Descriptive Statistics.

Variable	**Peso0**	**Peso1**	**Peso2**	**BMC**	**BMD**	**Tb.Sp**	**Tb.N**	**Longitud**	**GPSurf**	**GPVol**
Mean ($\bar{x}$)	105.70	174.44	302.45	0.28	143.26	0.33	3.13	38.80	40.43	5.60
ST (σ)	22.09	72.04	64.43	0.07	16.81	0.08	0.60	3.05	4.81	1.14
N (Obs.)	134	134	134	134	134	134	134	134	134	134

Note: Peso0, Peso1 and Peso2 in grams (g); BMC in mg; BMD in mg/cm^2; Longitud in milimiters (mm), and GPSurf in mm/mm^2; GPVol in mm^3.
Measure units of Tb.N and Tb.Sp are not specificated.

Table 2: Categorical Variables.

	Estudio			**Tiempo**		**Dosis (mg/day)**								
Options	**1**	**2**	**3**	**28**	**49**	**0**	**1**	**8**	**10**	**80**	**100**	**400**	**500**	**1000**
Total (N)	56	42	36	92	42	54	8	8	8	19	8	11	9	9
Percentage (%)	41.8	31.3	26.9	68.7	31.3	40.3	6.0	6.0	6.0	14.2	6.0	8.2	6.7	6.7

3.2 Statistical Methods

3.2.1 Principal Component Analysis

The first problem we faced as a group was dealing with the missing values and outliers. By performing the *Student's t-test*, we concluded that the missing values were not distributed randomly, even so we replaced them by the mean as it is stated in the instructions. The outliers were also changed by the mean.

Then, we made prior comprobations such as *Bartlett*'s sphericity test and analyze the similarity of each of our variables with a normal distribution.

In order to determine the number of Principal Components, we use the Rule of *Abdi et al* to select **three** principal components. As we can clearly see in the following histograph, the first three columns are the ones who variate the most between each other:

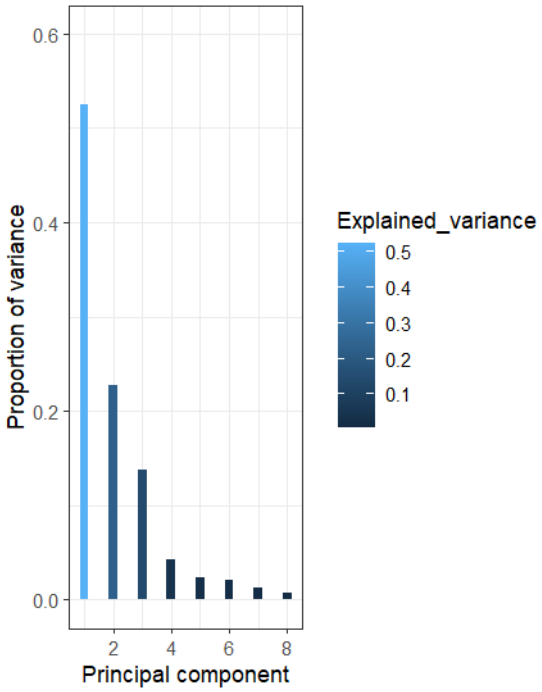


Figure 1: Caption

Without any doubt, the graphical output that reflects the best the behaviour betwen the variables is the following *biplot*, which relates the first and second principal components (PC1=Dim1, PC2=Dim2):

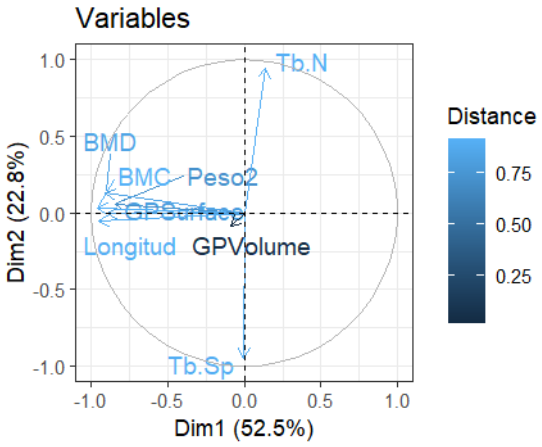


Figure 2

The arrows which are more horizontal are the ones who give more weight to the first principal component (PC1). Since they point to the left, we shall assume that the more negative values of PC1, the higher would be the values of each variable (*Longitud, BMC, BMD, GPSurface* and *Peso2*).

In contrast, the arrows which point up and down, almost completely vertical and in opposite directions, are the ones who explain the PC2. Both *Tb.N* and *Tb.Sp* values will be weighted by the same coefficient but with different sign (+ or -).

Finally the variable *GPVolume* represents by itself the whole PC3, that is why it is an small arrow that stays near the center of this plot, because it has nothing to do with PC1 and PC2.

3.2.2 Factor Analysis

As we have already done the Barlett test and prepared the variables that we want to reduce, we start visualizing the correlations to try to have some intuition on the number of latent variables.

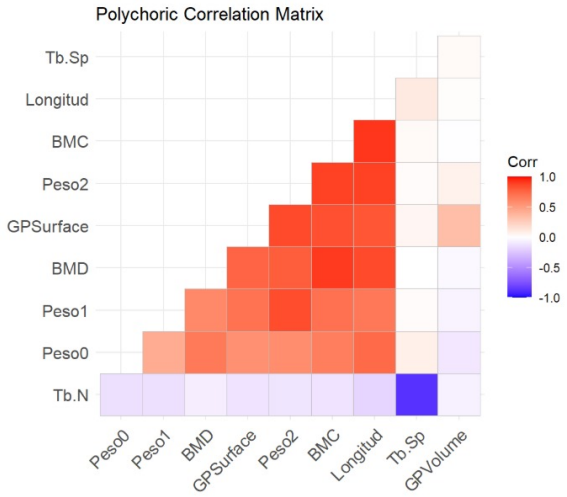


Figure 3: Visualization of the corrrelations

It easy to differentiate 3 groups in the matrix. Now we do the scree plot and the parallel analysis. Both suggest 4 factors, but when we fit the model with 4 factors using Varimax rotation and MLE, we see that it seems to be a better choice 3 factors excluding the initial and intermediate weights.

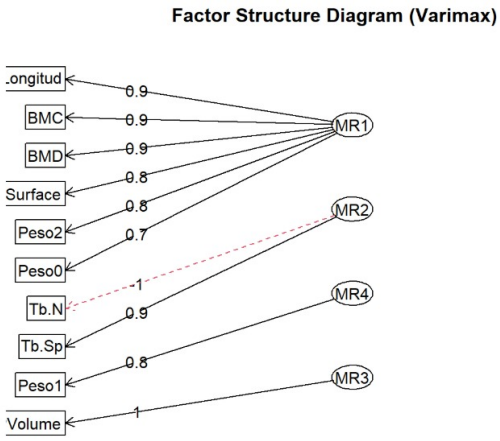


Figure 4: Diagram with 4 factors.

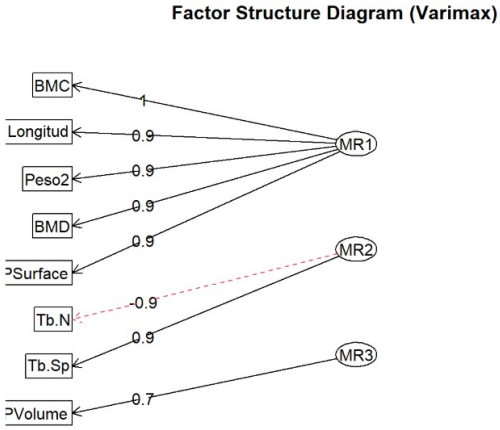


Figure 5: Final explanatory Diagram.

Finally we obtain the scores of each factor. Consider that a positive value in the first factor means a big growth. A negative value in the second factor means that has a good value of Tb.N, and bad value of Tb.Sp.. The third factor does not change as it has only one variable.

3.2.3 Discriminant Analysis

This step was crucial in order to obtain real conclusions about the study. We need to have in mind some of the statements we have previously achieved:

1. The more negative values of PC1, the higher would be the values of the variables *BMD, BMC, GPSurface, Longitud.*

2. The more positive values of Factor 1, the higher would be the values of the variables *BMD, BMC, GPSurface, Longitud.*

When performing the discriminant analysis, we plotted interesting graphics:

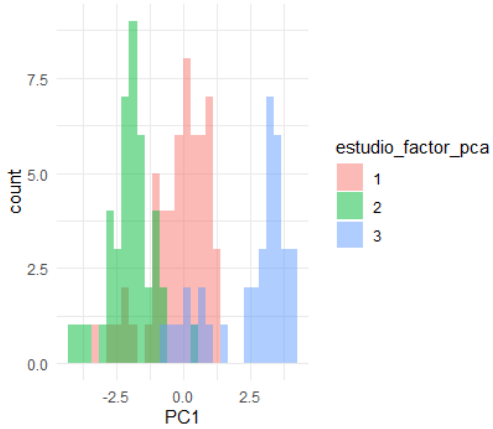


Figure 6: Values of PC1 based on the study.

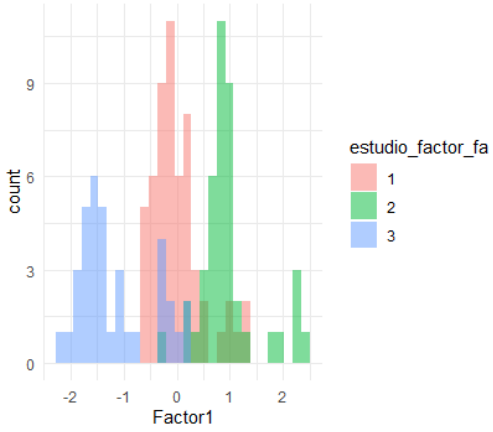


Figure 7: Values of Factor 1 based on the study.

From this we can draw some interesting conclusions: individuals who belonged to study 2 groups appear to have experienced greater bone progress (as their *BMD, BMC, GPSurface* and *Longitud* values are higher).

We decided to compare this information by obtaining these same graphs as a function of the dose, in order to get a solid conclusion.

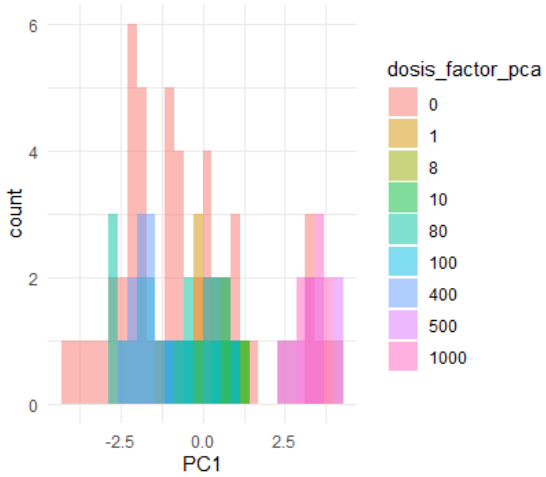


Figure 8: Values of PC1 based on the study.

It is clearly seen that extreme values of the dose (500-1000 mg/day) result in higher values of PC1, which translates into poorer bone quality. Intermediate drug doses appear to give better results.

Finally, performing a linear discriminant analysis, we could confirm our intuitions:

Table 3: Group means

Dosis	Peso2	BMD	Tb.N	Tb.Sp	GPVol	BMC	Long.	GPSurf
0	326.47	148.54	3.17	0.32	5.48	0.30	39.42	41.86
1	282.53	140.14	3.12	0.32	5.13	0.26	38.52	40.00
8	282.49	144.05	3.51	0.30	5.06	0.26	38.19	38.30
10	298.36	141.33	2.86	0.36	6.00	0.26	38.78	39.90
80	316.96	148.85	3.21	0.32	6.03	0.30	40.33	42.57
100	298.80	143.44	3.15	0.32	6.04	0.27	38.52	40.29
400	341.03	154.40	2.93	0.35	5.44	0.33	41.81	42.60
500	212.58	121.47	3.47	0.29	5.61	0.19	34.03	34.15
1000	212.76	111.60	2.48	0.42	5.77	0.17	34.00	33.84

Table 4: Coefficients of linear discriminants

	LD1	LD2	LD3	LD4	LD5	LD6	LD7	LD8
Peso2	0.003	0.002	-0.002	-0.007	-0.007	-0.034	-0.013	0.032
BMD	-0.085	-0.029	0.120	-0.062	0.073	-0.110	-0.025	-0.087
Tb.N	-1.540	-1.656	3.040	3.066	-3.981	0.977	-2.265	0.385
Tb.Sp	-4.928	-14.006	17.870	20.659	-33.759	2.226	-21.016	-6.576
GPVolume	0.053	-0.476	-0.336	0.820	0.688	-0.257	-0.164	0.004
BMC	41.218	47.644	-23.347	44.816	-6.641	29.943	19.505	-17.525
Longitud	-0.877	-0.960	-0.344	0.049	-0.090	0.126	0.236	0.342
GPSurface	-0.056	0.130	0.124	-0.386	-0.054	0.328	-0.241	-0.097
Prop. Trace	**0.5584**	**0.2061**	**0.1280**	**0.0357**	**0.0326**	**0.0260**	**0.0131**	**0.0000**

As we can observe, LD1 and LD2 explain a 75% of the total trace. Moreover, *BMC* plays an important role in both variables. This means that the administered dose greatly influences the development of *BMC*.
For the 500-1000 mg/day doses, we can appreciate a decrease in *BMC* compared to the group of rats that took a placebo.
Furthermore, for the group that took a dose of 1000mg/day, there was also an increase in the value of the variable *Tb.Sp*, that is, an increase in the space between the trabeculae of the tibia.

4 Results

1. High doses adversely affect rats.
2. The drug does not classify rats by their weight, but by the internal quality of their bones.
3. The most appropriate time and dose appear to be in study 2.

5 Conclusion

A significant findind in this study is the clear evidence of a decline in bone quality at a high dosage levels, more precisely, doses ranging from 500 to 1000 mg/day appear to have a toxic effect on individuals, leading on a deterioration of the bone architecture.

On the other hand, more moderate doses (between 8 and 100 mg/day) correlate with the best structural outcomes. But it is important to note that the results are not significantly differentiated from the placebo group.

Another important point is that the drug apparently works the same for all rats, regardless of their physical build - since weight variables are barely represented in the LDA variables. This

shows that the changes in bone architecture are caused by the drug itself, not by how much a mouse weighs or their constitution. In other words, the drug's effects, whether good or bad, do not depend on the physical size of the animal.

Regarding time, it is also matematically proved that the combination of a reasonable dose administered over a longer period of time yields better results.